

From a first answer to an audited one

Two real sessions with rstash, a local retrieval tool over a personal library of 500,000 passages. In each, the answer is criticised, the criticism is checked against the library, new sources come in only after review, and the system reports plainly what it can't settle.

October 2026 · every command was run and every output is quoted from a session on file

Contents

- 1. [The loop in one screen](#)
- 2. [Example 1: Hamiltonian Monte Carlo \(the full loop\)](#)
- 3. [Example 2: Can the Toyota bZ lock itself when I walk away?](#)
- 4. [What the two examples show](#)

The loop in one screen

rstash	Ask a question.
c: <critique>	Criticise the last answer. A multi-line paste is fine.
rstash --research	Look for outside sources (arXiv, OpenAlex) for points the library can't settle.
rstash --review	Approve or reject each source passage, with a reason. The answers are then re-checked.
rstash --gaps	Re-ask where an answer ignored evidence it was given.
rstash --verdicts	Override a verdict yourself, with a reason.
rstash --revise	Correct the critique where it claimed more than any source shows.

Behind c: , rstash takes four steps:

1. It splits the critique into points. Each point is a **claim** (you assert something) or a **question** (you ask or expect something).
2. It writes one **neutral follow-up question** per point. The question names the subject in full and never assumes your answer.
3. It answers each follow-up in **research mode**: the usual semantic search, plus the pages most relevant to the question and some model-written search terms, found by exact keyword search over whole pages.
4. An independent judge rules each point:

VERDICT	MEANING
answered	Your sources, cited in the answer, settle the point. A question can be settled <i>explicitly</i> or <i>by inference</i> .
partly supported	The sources back part of it. The report says what they establish, where the critique went further than any source, and any answer gap (evidence that was in front of the answer but went unused).
not settled	Your sources don't settle it. Research is the next step.

Each critique is a **session**, a folder holding the critique, the follow-up plan, the results with their history, any overrides, and a `REPORT.md` . Approved sources sit in a separate "approved additions" overlay next to the main index. Their citations carry `[added]` , so every answer shows which parts rest on sources you approved.

Example 1: "Hamiltonian Monte Carlo use in machine learning"

Step 1: the first answer

```
$ rstash
```

Question: Hamiltonian Monte Carlo use in machine learning

Hamiltonian Monte Carlo (HMC), also referred to as Hybrid Monte Carlo [algorithms_for_valuation.pdf p.168], is a Markov chain Monte Carlo (MCMC) method that utilizes Hamiltonian dynamics ... It is used to sample from complex, high-dimensional, and **multimodal** distributions ... HMC uses a simulation of Hamiltonian dynamics based on the gradient of the **log-likelihood** ... symplectic integrators ... ensure **stability** ...

The audit line looked perfect: answer 12 claims, 12 cited · citations match excerpt labels . But every claim being cited doesn't make the answer correct.

- A reviewer found one misleading claim (multimodality presented as a strength), one imprecision ("log-likelihood" instead of the log target density) and one major omission (the Metropolis accept/reject step).
- 8 of the 15 excerpts came from a single blog post. The textbook pages on HMC never reached the context.
- Against 12 checks drawn from that critique, the original answer passed **0–1 of 12**.

Step 2: criticise it

Paste the critique after c : at the prompt, or run a saved one from a file:

```
$ rstash --critique critiques/hmc.json 20261002T224615Z-rstash-acbf04f7
critique of 20261002T224615Z-rstash-acbf04f7 · 12 given points · writing neutral follow-up que
### How does Hamiltonian Monte Carlo perform when sampling from multimodal probability distrib
C1: answered The answer, citing Bou-Rabee & Sanz-Serna p.84, says HMC struggles with multim
because momentum randomization often lacks the energy or direction to cross the potentia
### Beyond stability, what roles do volume preservation and time reversibility play ...
C4: partly supported
answer gap: The answer never names the leapfrog integrator, although Okudo & Suzuki 2017
Betancourt 2017 p.37 state that leapfrog preserves volume and reversibility (with constan
critique done (150s): 7 answered · 5 partly supported · 0 contradicted · 0 not settled
```

Rust run, 5 Oct, on the full library plus the approved sources from steps 3 and 4.

Step 3: find outside sources

For points the library can't settle, rstash takes these steps:

- It writes search queries for arXiv and OpenAlex, and screens titles and abstracts.
- It downloads the best papers and picks their most relevant pages.
- It quotes the deciding sentences, checked **word for word** against the page. A quote that can't be found verbatim is dropped.

It searches for the part the critique claimed beyond the sources, not for the whole claim.

```
$ rstash --research
research 1 point(s): P2
P2: 13 hits → Zhang et al. (2023) · Du & Mordatch (2019), Implicit Generation and Generalizati
arXiv_1903.08689 p.3: PARTIAL We note that HMC [Neal, 2011] may be an even more efficient gr
arXiv_2110.10873 p.2: PARTIAL For sampling, most existing EBMs rely on the Langevin dynamics
6 new candidate(s) → rstash --review
```

Shown: the Rust research step on a test copy of an HMC session. The 14 papers in the approved set were found by the same procedure, in its Python prototype, on 3 and 4 Oct.

Step 4: decide

Each candidate appears with its critique point, its follow-up question, the claim being checked, and the passage highlighted in its page. A link opens the PDF at that page.

```
$ rstash --review
[6/6] P2 EBM image generation mostly uses Langevin dynamics (one-step HMC), not full HMC
claim to check: Energy-Based Model image generation primarily utilizes Langevin dynamics rat
```

```
source: Nie et al. (2021), p.2 (open) · web
... For sampling, most existing EBMs rely on the Langevin dynamics (LD) sampling ...
approve? (y/n[: reason], o=open PDF, q=quit): y: most existing EBMs rely on Langevin
recorded 3 approved, 3 rejected
overlay: 884 chunks from 15 approved source(s)
re-running the follow-up(s) for P2 with the updated overlay...
P2: partly supported → answered
answer gap now: none
```

Your reasons are part of the record. From the real HMC reviews:

```
APPROVE C4 p.2 provided the step size is kept constant
APPROVE C4 p.31 but does not say leapfrog integrator is symplectic, only that symplectic int
APPROVE C6d p.1 does not say it is default, only that it is popular (but if popular that imp
APPROVE C6d p.7 stronger wording as NUTS is described as the core Bayesian method
APPROVE C6b p.41 Approved with limits: Betancourt supports C6b only weakly ...
```

The approved set now holds **14 papers and 25 approved passages**, including:

- Neal 2011, Hoffman & Gelman 2014 (NUTS), Betancourt 2017 and Bou-Rabee & Sanz-Serna 2018;
- the Stan, PyMC3 and NumPyro papers;
- Song & Kingma 2021, Karimi et al. 2023 and Mangoubi et al. 2018.

Step 5: close answer gaps

Sometimes the evidence is in front of the answer and it doesn't use it. `--gaps` re-asks those follow-ups with a note that points at the unused excerpt. It adds no new facts, and the trace records the exact prompt.

```
$ rstash --gaps
### Beyond stability, what roles do volume preservation and time reversibility play ... · wit
C4: partly supported → answered
### What are the computational costs associated with gradient evaluations per leapfrog step ..
C6b: partly supported → answered
```

That took the session from 7 to **9 answered**. The three points left were all cases of the critique claiming more than any source shows: "usually" (C5), an explicit statement about Neal's funnel (C6c), and "default" in PyMC and NumPyro (C6d).

Step 6: correct the critique

Critiques can over-claim too. `--revise` shows each over-reaching claim, what the sources establish, and a draft rewrite. You accept the draft, write your own, or keep the original. The existing answers are then re-judged against the revised claims.

```
$ rstash --revise
C6c
original claim: Hamiltonian Monte Carlo struggles when the curvature of the target varies, s
draft revision: Hamiltonian Monte Carlo struggles in high-curvature neighborhoods, which can
revision? (a accept · e: <your wording> · k keep · q quit): e: Hamiltonian Monte Carlo struggl
curvature, which are characteristic of hierarchical models; Neal's funnel is a standard hier
re-judging the existing answers against the revised claims (no new answers)...
C5: partly supported → answered
C6c: partly supported → answered
C6d: partly supported → answered
```

POINT	ORIGINAL CLAIM	REVISED AND ACCEPTED
C5	EBMs are usually sampled with Langevin dynamics ...	Langevin dynamics, the one-leapfrog-step special case of HMC, is a popular, state-of-the-art sampler for training EBMs, alongside full HMC.
C6c	HMC struggles when curvature varies, such as Neal's funnel	HMC struggles in regions of high curvature, characteristic of hierarchical models ; Neal's funnel is a standard hierarchical test case.

POINT	ORIGINAL CLAIM	REVISED AND ACCEPTED
C6d	Stan, PyMC and NumPyro use HMC/NUTS as their default engine	Stan uses NUTS as its core inference method; PyMC and NumPyro implement it.

Review the drafts rather than accepting them blindly. In both HMC rounds the model's drafts over-corrected: one dropped a point the sources do support, and another softened "state-of-the-art" to "often". The review screen shows what the sources establish so you can judge.

Step 7 (optional): override a verdict

```
$ rstash --verdicts
P1 judge: partial · plain HMC is bad at multimodal targets ...
verdict? (a[: reason], r, k, q): a: the hedged version is what I meant
```

The report then reads *"answered by reviewer (the hedged version is what I meant; judge: partly supported)"*. Your view and the judge's view both stay visible.

Before and after

On multimodality, the first answer said:

It is used to sample from complex, high-dimensional, and multimodal distributions ...

The final follow-up answer, with sources:

HMC may not converge well when potential energy has multiple minima separated by large barriers [Bou-Rabee & Sanz-Serna 2018, p.84]. This is because momentum randomizations may fail to produce initial conditions with enough kinetic energy to cross potential energy barriers or may lack the correct direction to exit a current potential well [Bou-Rabee & Sanz-Serna 2018, p.84] ... For highly multimodal targets, HMC often does not beat RWM because the gradient within one mode is effectively ignorant of other modes [Mangoubi et al. 2018, p.1] ... HMC can be used in conjunction with tempering techniques ...

Original answer	0-1
c: first pass	7
after --gaps	9
after --revise	12
answered	partly supported
points out of 12	

No unsourced text was pasted into an answer. Every improvement came from a source in the library or one approved in review, and every step left a record.

Example 2: "Can the Toyota bZ lock itself when I walk away?"

A practical question about a 729-page owner's manual. Two Nissan Ariya manuals share the same index.

Step 1: ask

```
$ rstash
Question: can the toyota bz automatically lock the doors when the driver walks away from the c
```

Yes, the Toyota bZ has a "close & lock (Walk-Away) function," though this setting must be customized at a Toyota dealer [2027-Toyota-bZ.pdf p.175]. To use this function, the user must:

- 1. Close all doors except the back door while carrying an electronic key.
- 2. Press the switch on the lower part of the back door, which triggers a specific buzzer and puts the function into standby.

3. Move away from the back door while the buzzer is sounding [p.175].

... all doors other than the back door will lock, and the back door will lock once it has closed [p.176].

That's an interesting find: a walk-away function exists, but it starts at the back door. Is it really automatic?

Step 2: criticise it

c: this still requires manual button pressing by the user (and being at the back door); The Ariya does not have automatic locking of all doors by walking away from the car. It seems the bZ does not?

- P1 claim The Toyota bZ 'close & lock (Walk-Away) function' requires the user to manually press the button on the back door
- P2 claim The Nissan Ariya enables automatic locking of all doors by walking away from the car
- P3 question Does the Toyota bZ have a fully automatic walk-away door locking feature that does not require manual button pressing?

POINT	VERDICT	WHAT RSTASH FOUND
P1	answered	"requires the user to press the switch on the lower part of the back door while carrying an electronic key and having all other doors closed" (p.175)
P2	partly supported	The Ariya has a "Walk away lock function" (2024 manual p.277), but no excerpt describes what it does.
P3	answered, inferred	No. "The Toyota bZ's close & lock (Walk-Away) function must be started by pressing the switch on the lower part of the back door, and it must be enabled by a Toyota dealer."

P3 shows why an answer settled by inference is useful. The manual never says the bZ cannot lock itself, because manuals describe what a car *does*. rstash reports the conclusion the pages support, labels it as inferred, and shows the reasoning.

Step 3: read "partly supported" carefully

For P2, the follow-up retrieved only the Ariya manual's **contents entry** (p.277), not the page that describes the feature. The verdict reflects a retrieval miss inside the library, not missing knowledge. A narrower question finds the page:

\$ rstash "What does the Nissan Ariya walk away lock function do?"

The Nissan Ariya walk away lock function automatically locks the vehicle when you walk away from it with the Intelligent Key, with an operation range of approximately 7 ft (2 m) [2026-Nissan-Ariya-TypeD-OM-1st.pdf p.274]. When the vehicle is locked via this function, the hazard indicator lights flash twice ... This function is disabled by default, but it can be enabled through the vehicle information display ...

To record that conclusion in the session, `rstash --verdicts` lets you accept P2 with a reason, for example `a: Ariya manual p.274 describes it; asked directly`.

The answer

- **Toyota bZ:** walk-away locking exists, but it is **not fully automatic**. A Toyota dealer has to enable it, and each time you start it by pressing the switch under the back door (pp.175–176).
- **Nissan Ariya:** **fully automatic** once switched on in the vehicle information display, where it is off by default. It locks when you walk more than about 7 ft (2 m) away with the key (2026 manual p.274).

This matches what owners report: the bZ unlocks itself as you approach but doesn't lock as you leave, while the Ariya does both. The bZ's regenerative-braking level follows the same pattern. It isn't remembered between drives, a conclusion settled by inference in an earlier session from the manual's cancellation rules (pp.244–245).

What the two examples show

- **The critique step corrects retrieval by itself.** Naming the subject in every follow-up and adding exact keyword search over whole pages brought in pages the first, naive question missed.
- **"Not settled" and "partly supported" are informative.** They say exactly what is missing: a source (research), unused evidence (--gaps), a claim that went too far (--revise), or a page the library has but didn't retrieve (ask more narrowly).
- **You decide what counts as true.** New sources enter only through review, with your reasons on record. Overrides are labelled as yours, and the critique itself can be corrected.
- **Every result is traceable.** Each answer, judgement and decision is written to the session folder and the audit log.

rstash is a pure-Rust retrieval tool: GPU-exact search over SEC filings, PDFs, transcripts, web clippings, image descriptions and Kindle highlights, with a local or cloud language model for answers and a tool-less Claude call as the independent judge.

Markdown version for GitHub: `rag_audit/rstash_improvement_lab/WALKTHROUGH.md` . Sessions on file: `s13_r6/audit/critiques/20261002T224615Z-rstash-acbf04f7_20261005T191749/` (HMC) and `~/cache/rag_audit/critiques/20261005T194239Z-rstash-c63b6d90_20261005T194723/` (Toyota).